

Design and Implementation of 3G-324M

- An Event-Driven Approach

Bo Han, HaoHuan Fu, Ji Shen, Pui-On Au, Weijia Jia
Department of Computer Engineering and Information Technology
City University of Hong Kong
83 Tat Chee Avenue, Kowloon, Hong Kong
Email: hanbo00@cityu.edu.hk

Abstract—3G-324M is a derivative of the ITU-T H.324 standard for low bit rate multimedia communication. H.324 and several mobile specific annexes are usually referred to as H.324M (M stands for mobile). Then the 3GPP adopted the H.324M with some modifications in codecs and error handling requirements to create the 3G-324M standard. So 3G-324M is an “umbrella standard” referencing other standards which specify mandatory and optional video and audio codecs, the messages to be used for call set-up, control and tear-down and the way that audio, video, control information and other data are multiplexed and demultiplexed. In order to support enhanced and delay sensitive video services, 3G video phones as well as network equipments are required to support 3G-324M. In this article, we describe the design and implementation of 3G-324M protocol stack in details. In order to implement this complicated protocol stack, we present an approach based on event-driven programming. We have achieved efficient implementations for both Intel Pentium Processor and ARM Processor, which prove the feasibility of our approach. The performance of our implementation is also satisfactory.

Keywords—3G-324M protocol stack; Event-driven approach.

I. INTRODUCTION

The third generation (3G) International Mobile Telecommunications 2000 (IMT-2000) has started to be deployed, and its service has just begun in some countries. However, even before its deployment, its enhancement activity had already started and has been actively pursued in the Third Generation Partnership Project (3GPP [10]). Furthermore, in various fora and organizations for wireless research, such as the Wireless World Research Forum (WWRF [11]) and International Telecommunication Union (ITU), there have been active discussions about 3G systems. According to [1], the number of mobile subscribers over the whole world increased from 300 million in 1997 to 800 million in 2001. It is predicted that by 2010 there will be 1700 million subscribers worldwide. Such growth is driven by increasing user expectations for mobile services. Users will demand more multimedia services, and expect new applications and services that are ubiquitous and customized to individual needs. In order to meet rising expectations, telecommunication services will be integrated and converged in the future. For example, digital broadcasting and commercial wireless services will be integrated to create new value-added services and bring enormous benefits to both end users and service providers.

Video phone technology and products are not new to the marketplace. However, initial products often required the use of ISDN phone lines to carry video phone signals at data rates higher than what is available on common analog telephone lines. Then the gradually mature modem technology allowed analog telephone lines to be used for video phone calls. And there have been a number of first generation video phone products that connect over analog telephone lines. Only recently has mobile/wireless technology advanced enough to allow wireless channels to be used for video/audio data transmission. ITU-T H.324 [2] is an umbrella protocol defined by ITU to enable multimedia communication over low-bit rate terminals (in the following, we will drop “ITU-T” from the prefix of standard names for simplicity). H.324 and its several mobile specific annexes are usually referred to as H.324M (M stands for mobility). 3GPP has adopted the H.324M with some modifications in codecs and error handling requirements to create the 3G-324M standard for circuit-switched 3G networks. In order to support the enhanced and delay sensitive video services among heterogeneous terminals, 3G video phones or terminals are required to support 3G-324M protocol stack.

We have developed and implemented an efficient mobile multimedia transmission protocol stack based on 3G-324M standards in our 3G-AnyMobile project with some Chinese mobile companies. In such implementation, the performance and quality of service are critical to the successful execution of the entire system. This paper discusses some efficient approach and experience in the implementation of 3G-324M protocol stack. Specifically, we discuss the event-driven approach for the overall information exchange in facing hundreds of messages/states. Our implementation has been tested in a realistic heterogeneous 3G communication environment in some China industries for transmission of real-time video, audio and data and its performance is satisfactory.

The rest of the paper is organized as follows. In Section II, we introduce the background of H.324/3G-324M. Section III describes our design and implementation of 3G-324M protocol stack using Event-Driven Programming. We point out future directions and summarize major results in Section IV and Section V.

The work is supported by Research Grant Council (RGC) Hong Kong, SAR China, under grant nos.: CityU 1055/00E and CityU 1039/02E.

II. FROM ITU-T H.324 TO 3G-324M

A. The ITU-T H.324 Standard

ITU-T H.324 is a standard that provides a foundation for interoperability and high quality video, voice and data services based on phone calls. H.324 specifies a common method for video, voice and data to be shared simultaneously through modem connections operating over PSTN. Once a modem connection has been established, the H.324 standard specifies how digital video and voice compression technologies are used to convert sounds and facial expressions into digital signals. It also defines how these signals are compressed to fit within the data rate allowed by analog phone lines and modem connections. The standard describes terminals for low bit-rate multimedia communication. It is the first standard to specify interoperability over a single analog line. This provides a foundation for market growth and means that the video phone products will be able to talk to one another. Support for each media type (voice, data, or video) is optional, but if supported, the ability to use a specified common mode of operation is required, so that all terminals supporting that media type can communicate with each other. It allows more than one channel of each type to be used, and makes use of the logical channel signaling procedures of H.245, in which the content of each logical channel is described when the channel is opened. Procedures are provided for expression of receiver and transmitter capabilities, so transmissions are limited to what receivers can decode, and receivers may request a particular desired mode from transmitters.

B. The 3G-324M Standard

3G-324M [6][7][8] is a derivative of ITU-T H.324 standard for low bit rate multimedia communication. 3G-324M specifies

the use of a dedicated 64kbps bearer to transmit multiplexed voice, video as well as control and signaling information. It is an “umbrella standard” referencing other standards which specify mandatory and optional video and audio codecs, the messages to be used for call set-up, control and tear-down (H.245 [3]) and the way that audio, video, control information and other data are multiplexed and de-multiplexed (H.223 [4]). Figure 1 below shows the functional components of a generic 3GPP multimedia terminal. Note that the differences between 3G-324M and H.324M mainly lie in codecs (voice by AMR-Adaptive Multi Rate Speech Codec, and video by H.263 or MPEG-4) and error handling requirements (H.223 Annex A and Annex B as mandatory). Therefore, 3G-324M inherits H.324M basically but must use AMR for speech codec. The AMR speech coder consists of the multi-rate speech coder, a source controlled rate scheme including a voice activity detector, a comfort noise generation system, and an error concealment mechanism to combat the effects of transmission errors and lost packets.

H.245: The control protocol--H.245 specifies syntax and semantics of terminal information messages as well as procedures for in-band negotiation at the start of or during communication. The messages cover receiving and transmitting capabilities as well as mode preference from the receiving end, logical channel signaling, control and indication. The message syntax is defined using ASN.1 notation. For transmission, ASN.1 formatted data is transformed into bit-stream based on ASN.1 encoding rules, Packed Encoding Rules (PER) [5]. H.245 has been defined to be independent of the underlying transport mechanism, but is intended to be used with a reliable transport layer, that is, one that provides guaranteed delivery of correct data.

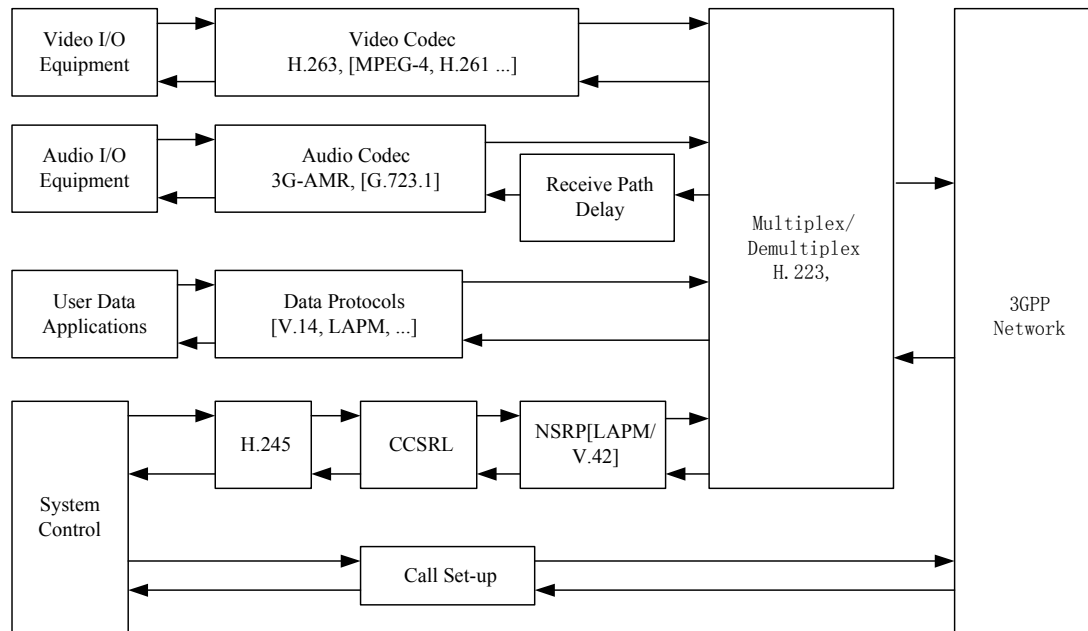


Figure 1. 3G-324M Protocol Stack

H.223: The multiplexing protocol consists of the Multiplex (MUX) Layer and Adaptation Layer (AL) -- AL is actually an interface for upper-layer applications, and it deals with different source separately. The MUX layer may perform the actual multiplexing. In this layer, data traffic from different sources can be multiplexed into one packet according to some rules which are negotiated by two terminals at the beginning of communication. These rules are described in the forms of multiplex tables. This protocol provides low delay and low overhead by using segmentation and reassembly, and by combining information from different logical channels into a single packet.

III. THE DESIGN AND IMPLEMENTATION: AN EVENT-DRIVEN APPROACH

Our implemented protocol stack is fully compatible with 3G-324M, making it a popular choice for deployment for the following applications: (1) 3G-324M wireless handsets for TD-SCDMA/WCDMA/CDMA2000 networks; (2) 3G-324M multimedia gateways to connect 3G wireless and landline networks; (3) Real-time multimedia streaming solutions over 3G-324M.

A. Event-Driven Programming

To coordinate the different parts of the 3G-324M protocol stack efficiently, our implementation for 3G-324M applies event-driven programming which is characterized by triggering the execution of protocol in an arbitrary order, rather than in a pre-determined order such as in a sequential control procedure [9]. This approach fits to the behaviors of network protocols as their actions may be unpredictable. Based on event-driven programming, it is easy to control the software architecture and maintain a cooperation of the protocol stack in the overall system. Figure 2 shows the event-driven based system control in the protocol stack. H.245 module and H.223 module use the common buffer to swap control information exchanged between peer terminals. We will discuss the Event Generator and the Event Dispatcher later.

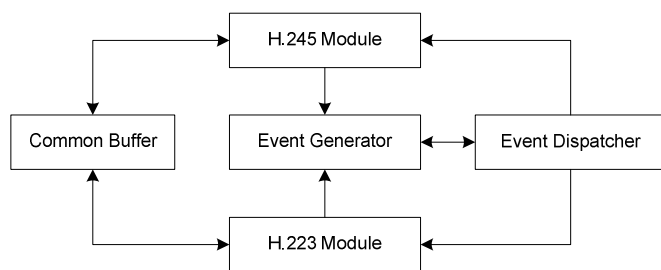


Figure 2. Event-Driven Based System Control

B. Overview of 3G-324M Protocol Stack

Although events are not defined explicitly in 3G-324M specification, there are generally three types of events in 3G-324M: messages, primitives and timeout events. Message events occur when the terminal receives control messages from another terminal; primitive events happen when there is an

```

typedef struct _H324Event
/* Definition of H324 Event */
{
    int nEventID;
    int nEventType;
    void * pEvent;
    /* pointer of the detailed event */
    struct _H324Event * next;
    /* pointer of the next event */
} H324Event;

```

Figure 3. C language struct of H324Event

```

H324_Listen_Event()
{
    while (!bCallEnd)
    {
        H324_Timeout();
        H324_Recv_Message();
        H324_Recv_Primitive();

        if (bHasNewEvent)
            H324_Dispatch_Event();
    }
}

```

Figure 4. The main system control function

interaction between 3G-324M and its upper layer applications; and timeout events occur when timer expires in 3G-324M protocol stack. Figure 3 shows the data structure of *H324Event* in C language. In the event-driven control approach, two core parts are defined as Event Generator and Event Dispatcher. Event Generator generates corresponding events based on the current state and takes care of the maintenance of event queue and timer chain. Event Dispatcher calls related handling procedures based on the finite state machine definitions and it may also call Event Generator to generate other events. Our implementation uses event generator and dispatcher to coordinate H.245 and H.223 protocols to send/receive multimedia data as well as the control messages. Thus, the event-driven mechanism is executed across the entire 3G-324M protocol stack. Figure 4 gives the main functions which make up of system control in C language.

C. Event Processing

3G-324M defines a general message type *Multimedia-SystemControlMessage* (MSCM). Four types of special messages are further defined in MSCM as *request*, *response*, *command* and *indication*. A *request* message results in a specific action and requires an immediate response. A *response* message responds to a request message. A *command* message requires an action but no explicit response. An *indication* message contains information that does not require action or response. Messages with various types are transformed into MSCM for uniform processing.

In 3G-324M, Signaling Entity (SE) is referred to as a procedure that is responsible for special functions. It is designed as a finite state machine and changes its current state upon reaction to an event occurrence. There are ten types of SEs defined in H.245 (Annex C) as shown in Table 1. Taking *Master Slave Determination Signaling Entity* (MSDSE) as an example, it is a procedure for master-slave determination between two peer terminals. And it has three states: *IDLE* (no master-slave determination procedure is initiated), *OUTGOING_AWAITING_RESPONSE* (the local MSDSE has requested a master-slave determination procedure, and waits for a response from the remote MSDSE), *INCOMING_AWAITING_RESPONSE* (the local MSDSE has received a request of master-slave determination from the remote MSDSE and sent back an acknowledgement, waiting for a response from the remote MSDSE). Figure 5 shows the event processing of MSDSE in 3G-324M protocol stack. Consider a new message *MasterSlaveDetermination* (MSD) is firstly received from the peer terminal, H.223 module demultiplexes and

passes it to *Event Generator*. *Event Generator* generates an event for the message and *Event Dispatcher* recognizes it as MSD, and invokes MSDSE to handle it. Assume that the current state of MSDSE is *IDLE*, then the message event MSD triggers *Event Dispatcher* to execute the procedure *MSD_Msg_Proc*. Two parameters in the MSD message, *terminalType* and *statusDeterminationNumber* are used to determine the master-slave state. If the determination succeeds, *MSD_Msg_Proc* generates an *MSDAck* message and a primitive. *MSDAck* is then sent to the peer terminal through H.223 module to notify the success, while the primitive is sent to the system control to indicate the success. If the determination fails, *MasterSlaveDeterminationReject* (MSDRej) message is generated and sent to the peer terminal to notify the failure. Then *MSD_Msg_Proc* switches the current state of MSDSE to *OUTGOING_AWAITING_RESPONSE* for waiting response. Once an event procedure is finished, H.245 module will wait for the next event.

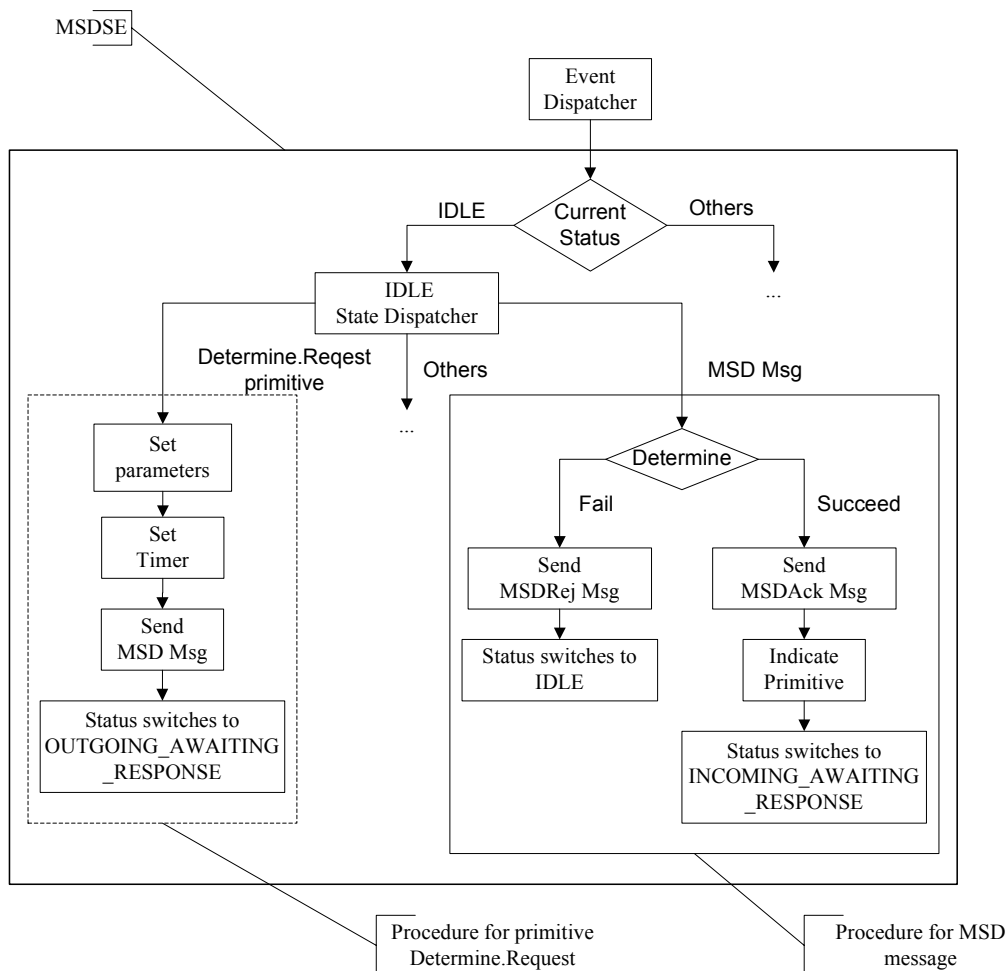


Figure 5. The event processing of MSDSE in 3G-324M protocol stack

B-LCSE	Bi-directional Logical Channel Signaling Entity
CESE	Capability Exchange Signaling Entity
CLCSE	Close Logical Channel Signaling Entity.
LCSE	Logical Channel Signaling Entity
MLSE	Maintenance Loop Signaling Entity
MSDSE	Master Slave Determination Signaling Entity
MTSE	Multiplex Table Signaling Entity
MRSE	Mode Request Signaling Entity
RMESE	Request Multiplex Entry Signaling Entity
RTDSE	Round Trip Delay Signaling Entity

TABLE I. TEN TYPES OF SIGNALING ENTITY

IV. FUTURE WORK

We have discussed the point-to-point multimedia transmission for the 3G terminals. With the key technology used as mentioned before, our implementation has gained a satisfactory performance. However, the multipoint multimedia transmission application such as video conferencing has not been addressed. Video conferencing is a technology that has been emerging over the past thirty years, motivated by how much of our face-to-face communication is visual, or nonverbal. Recent advances in media compression technology along with shrinking component size and cost are making digital video feasible. 3G-324M terminals may be used in multipoint configurations via interconnection through Multipoint Control Units. We are going to extend our implementation into a platform for video-conferencing in the 3G-AnyMobile project.

V. CONCLUSIONS

3G-AnyMobile is an ongoing research project that implements the 3G-324M protocol stack. We have finished the design and implementation on both Intel Pentium Processor and ARM Processor. In our implementation, event-driven

approach is used for efficient control of overall protocol stack in dealing with the complex states and control transfer for the multimedia data transmission. Our implementation has been tested in realistic heterogeneous 3G communication environment for transmission of real-time video, audio and data and its performance is satisfactory.

ACKNOWLEDGMENT

The authors want to thank Lin Lidong for the implementation of EGolay code, and other MMU research group members who helped us to implement X.691 and H.245 messages generation and transformation.

REFERENCES

- [1] ITU-R PDNR WP8F, "Vision, Framework and Overall Objectives of the Future Development of IMT-2000 and Systems beyond IMT-2000," 2002.
- [2] ITU-T Recommendation H.324, "Terminal for low bit rate multimedia communication", March 2002.
- [3] ITU-T Recommendation H.245, "Control protocol for multimedia communication", July 2003.
- [4] ITU-T Recommendation H.223, "Multiplexing protocol for low bit rate mobile multimedia communication", July 2001.
- [5] ITU-T Recommendation X.691, "Information technology – ASN.1 encoding rules – Specification of Packed Encoding Rules (PER)", Dec. 1997.
- [6] 3GPP TS 26.110 V5.0.0, "Codec for Circuit Switched Multimedia Telephony Service; General Description", June, 2002.
- [7] 3GPP TS 26.111 V5.0.0, "Codec for Circuit Switched Multimedia Telephony Service; Modifications to H.324", June, 2002.
- [8] 3GPP TR 26.911 V5.1.0, "Codec for Circuit Switched Multimedia Telephony Service; Terminal Implementor's Guide", April, 2003
- [9] George C. Philip, "Software design guidelines for event-driven programming," The Journal of Systems and Software, 41(2): 79-91, 1998
- [10] The 3rd Generation Partnership Project, <http://www.3gpp.org>
- [11] Wireless World Research Forum, <http://www.wirelessworld-research.org>